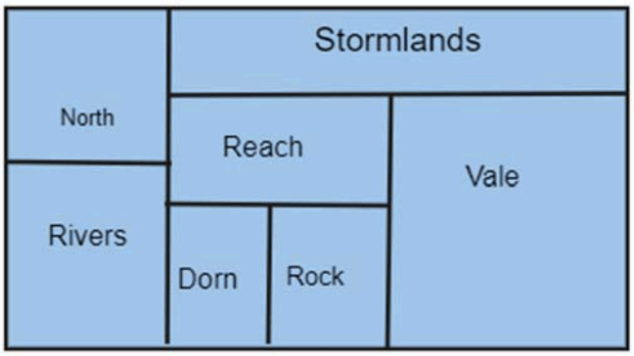
Practice problem (CSP)

Consider the following map of a Country where borders among the seven states have been marked. You need to color each of these divisions using three colors (Red/Green/Blue) such that two adjacent states do not receive the same color.
Now formulate the problem as CSP, Show the Constraint Graph, and solve the problem must applying both Minimum Remaining Values(MRV) and Least Constraining Value(LCV) heuristics. **[2.5+2.5]**



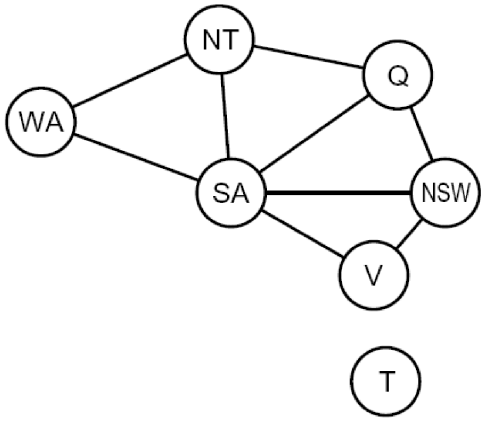
6. Consider the following map of Bangladesh where borders among the eight divisions have been marked. You need to color each of these divisions using three colors (Red/Green/Blue) such that two adjacent divisions do not receive the same color.
   Now **formulate** the problem as CSP, Show the **Constraint Graph** and solve the problem **must applying** both Minimum Remaining Values(MRV) and Least Constraining Value(LCV) heuristics. **[5]**



For the given graph you have color all the nodes with red/green/blue so that the adjacent nodes don't contain same color:

i) Define the variables and domain for this problem

ii) Solve the problem as a CSP following **Minimum Remaining Value (MRV)** and **Least Constraining Value (LCV)** heuristics.



Sample Solution

i) Variables: WA, NT, Q, NSW, V, SA, T
Domain: {red, green, blue}

Step 1: Using MRV- all nodes have same number of remaining value (red, green value)
Break tie using degree heuristic, and choose SA.
Using LCV- all colors have the same impact, so choosing anyone (red).
SA←Red

Step 2: Using MRV- (WA, NT, Q, NSW, V) all have same number of remaining value (green, blue)
Break tie using degree heuristics, and then NT, Q, NSW have the same number of edges.
So, we can choose any, now choosing NT.
Using LCV- all colors have same impact, so choosing anyone (green)
NT←green

Step 3: Using MRV- (WA,Q) both have one color remaining.
Break the tie with using degree heuristic, and choose Q.
Q←blue (only blue is the option)

Step 4: Using MRV- (WA,NSW) both have one color remaining.
Break the tie with using degree heuristic, and choose NSW.
NSW←green (only green is the option)

Step 5: Using MRV- (WA,V) both have one color remaining.
The both have same degree (same number of edge), so choose any.
WA←B
V←B

Step 6: T is independent (no edge) so put any color
T←red

6. A team of six employees is planning the seating arrangement for a new office layout. They are to be seated in any of the four departments – Finance, Marketing, HR, or IT. The following constraints apply:

- Employee A and Employee D want to be in the same department.
- Employee C prefers to work in the Finance department.
- Employees B and E are not on good terms and want to work in different departments.
- Employee D dislikes the Marketing department and will not sit there.
- Employee E is open to working only in HR or Finance
- Employee F must work in the same department as Employee C but in a different department from Employee E.

Formulate the problem as a CSP, clearly defining the variables, domains, and constraints, and solve the problem by applying both **Minimum Remaining Values(MRV) and Least Constraining Value(LCV) heuristics.** **[6]**

1. Four friends, Ryomen (R), Satoru (S), Yuji (Y), and Mahito (M) have come to a restaurant near their university. The restaurant serves Chicken Burger (C), Beef Burger (B), Kashmiri Naan (K), and Paratha (P). Their preferences are as follows:

i. Ryomen does not like Paratha.
ii.. Satoru and Yuji would like to grab a bite of each other's food, so they will order different dishes.
iii. Yuji likes burgers, so he will order a Chicken Burger or a Beef Burger.
iv. Ryomen likes to copy Satoru, so he will order the same dish as Satoru.
v. Mahito will not order Kashmiri Naan because he had them earlier.
vi. Mahito is obsessed with being unique, so he will not order the same dish as anyone else.

Based on these constraints, answer the following questions:

a. Draw the constraint graph for this CSP. **[2]**
b. Solve the CSP using **MRV and LCV**. In case of any ties, choose the **alphabetically smaller** option. **[6]**